HYBRIDIZATION

Definition: The phenomenon of mixing up of atomic orbitals of similar energies and formation of equivalent number of entirely new orbitals of identical shape and energy is known as "hybridization" and the new orbitals so formed is called as "hybrid orbitals".

Important points for understanding the hybridization:

(i) The number of hybrid orbitals generated is equal to the number of pure atomic orbitals that participate in hybridization process.

(ii) Hybridization concept is not applicable to isolated atoms. It is used to explain the bonding scheme in a molecule.

(iii) Covalent bonds is polyatomic molecules are formed by the overlap of hybrid orbitals or of hybrid orbitals with unhybridized ones.

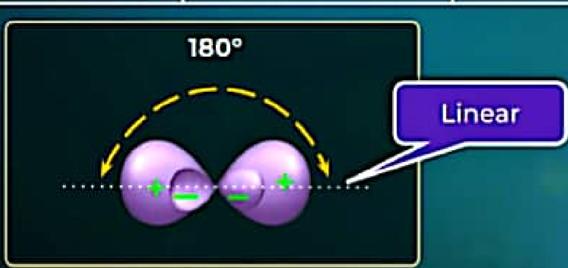
⇒ Different Types of Hybridisations:

There may have different types of hybridisation of orbitals. Such as:

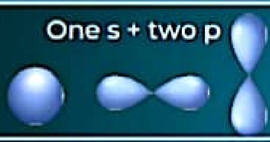
	Structures:
1. sp^3 hybridisation.	Tetrahedral
2. sp^2 "	Plane trigonal
3. sp "	Linear
4. sp^2d "	Square Planar
5. sp^3d "	Trigonal bipyramidal
6. sp^3d^2 "	Octahedral
7. sp^3d^3	Pentagonal bipyramidal
8. d^2sp^3 or sp^3d^2	Octahedral.

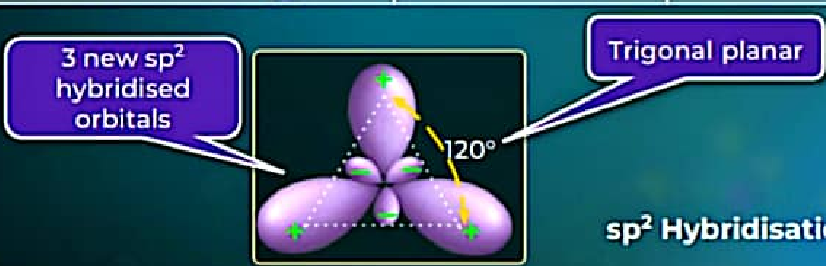
sp Hybridisation

Participating atomic orbitals	Number of hybridised orbitals	Hybridisation
One s + One p 	2	sp



sp^2 Hybridisation

Atomic orbitals participating in hybridisation	Number of hybridised orbitals	Hybridisation
One s + two p 	3	sp^2



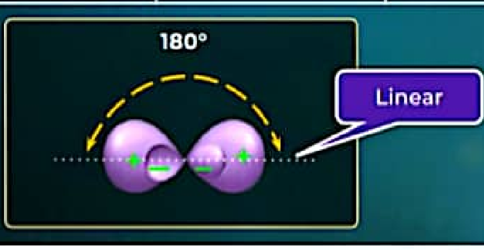
sp^2 Hybridisation

sp^3 Hybridisation

Atomic orbitals participating in hybridisation	Number of hybridised orbitals	Hybridisation
One s + three p 	4	sp^3

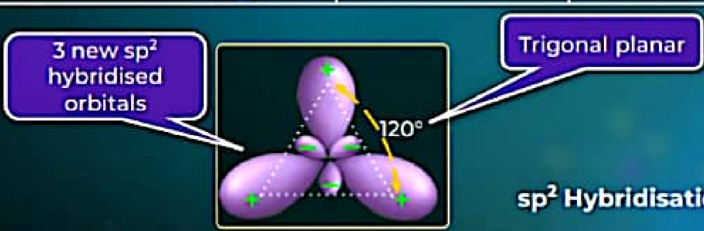
sp Hybridisation

Participating atomic orbitals	Number of hybridised orbitals	Hybridisation
One s + One p	2	sp



sp^2 Hybridisation

Atomic orbitals participating in hybridisation	Number of hybridised orbitals	Hybridisation
One s + two p	3	sp^2

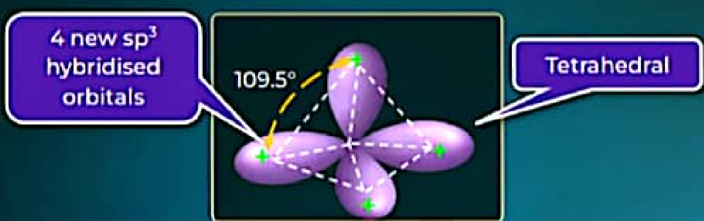


sp^2 Hybridisation

sp^3 Hybridisation

Atomic orbitals participating in hybridisation	Number of hybridised orbitals	Hybridisation
One s + three p	4	sp^3

sp^3 Hybridisation



Did you Know?

Hybrid Orbital

Actual Shape

Shape used for representation

Naming of Hybrid Orbitals

On the basis of atomic orbitals participating in hybridization:

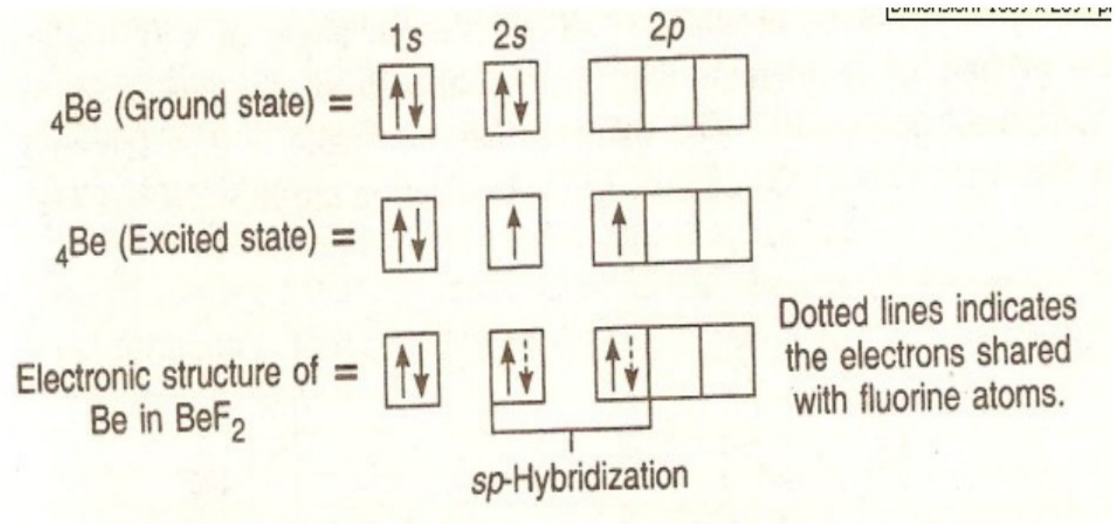
s-orbital + 'n' p-orbital + 'm' d-orbital

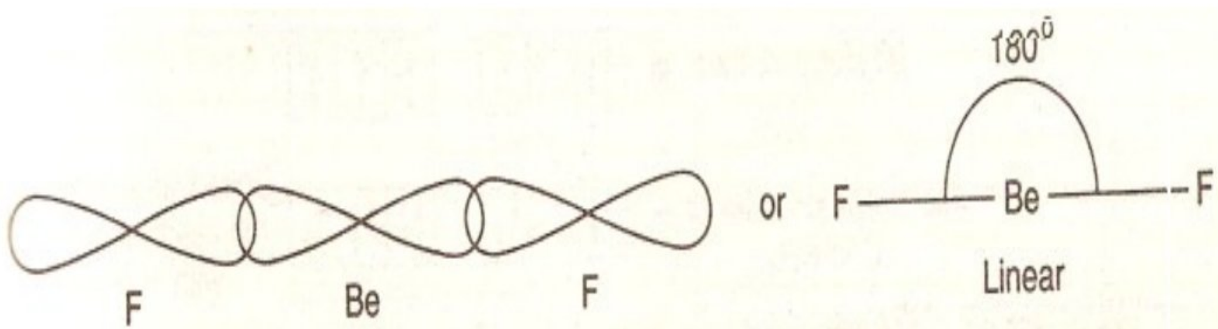
↓

sp^nd^m hybrid orbital

HYBRIDIZATION

Structure of BeF_2 Molecule

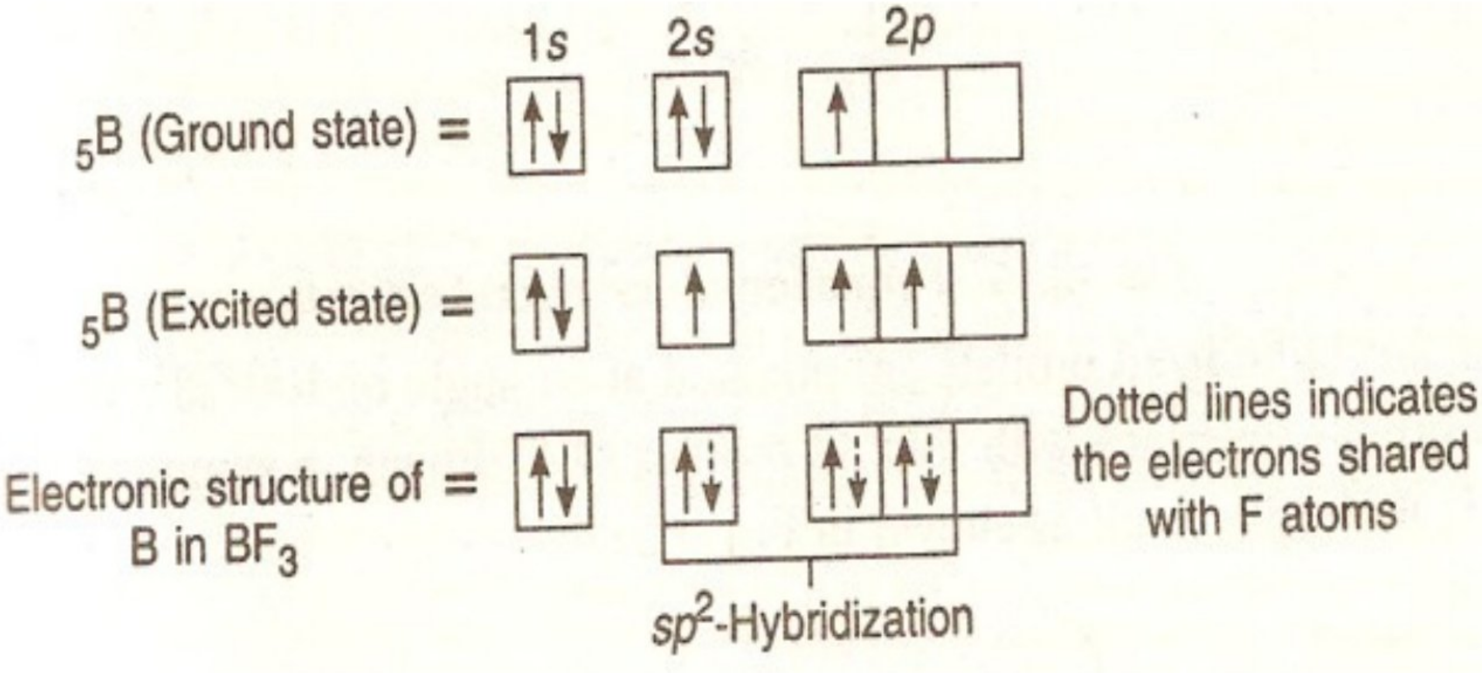




HYBRIDIZATION

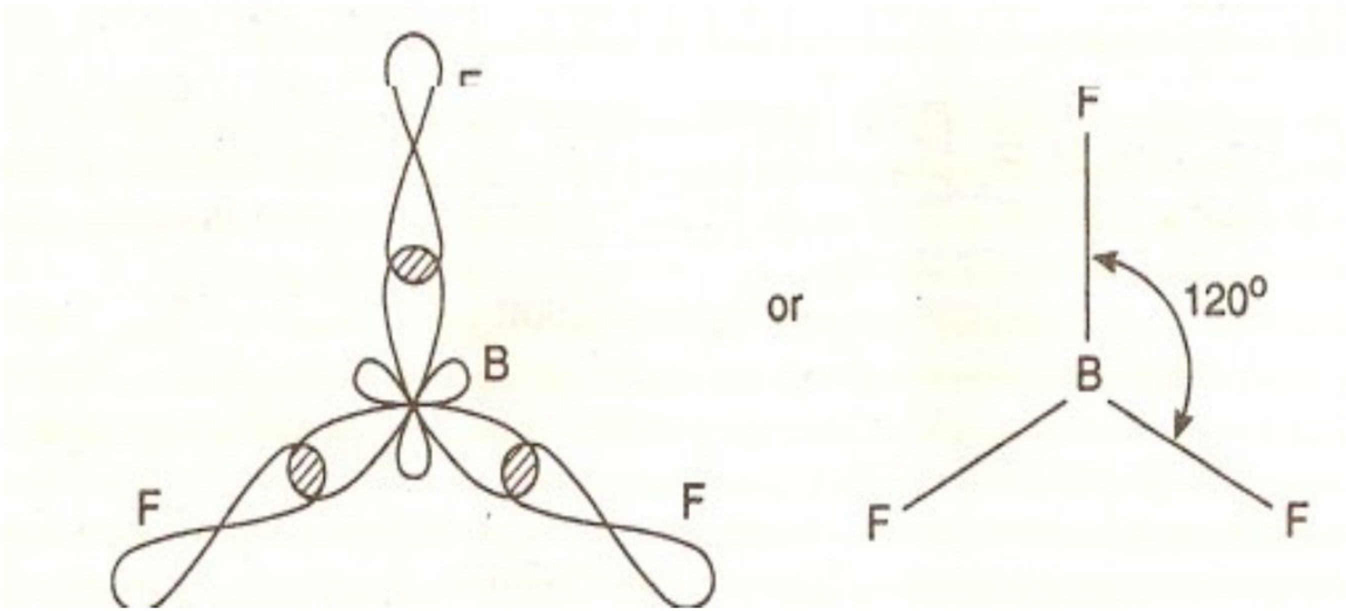
sp^2-hybridization: The combination of one s and two p-orbitals to form three hybrid orbitals of equal energy is known as sp^2-hybridization.

Example : BF_3 Molecule.



HYBRIDIZATION

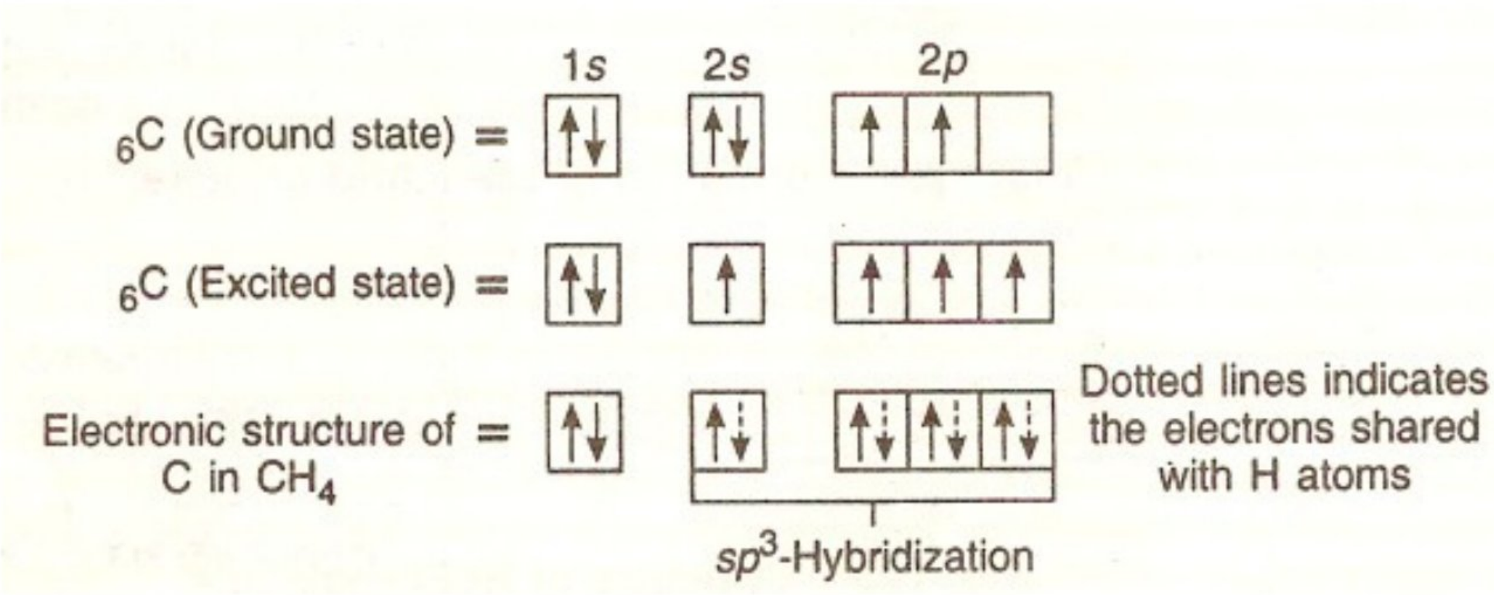
These sp2 hybridized orbitals are oriented at an angle of 120°. When three sp2 hybridized orbitals of B overlaps with three p-orbitals of fluorine, three σ -bonds are formed with bond angle 120°. The shape of BF3 molecule is thus trigonal planar



HYBRIDIZATION

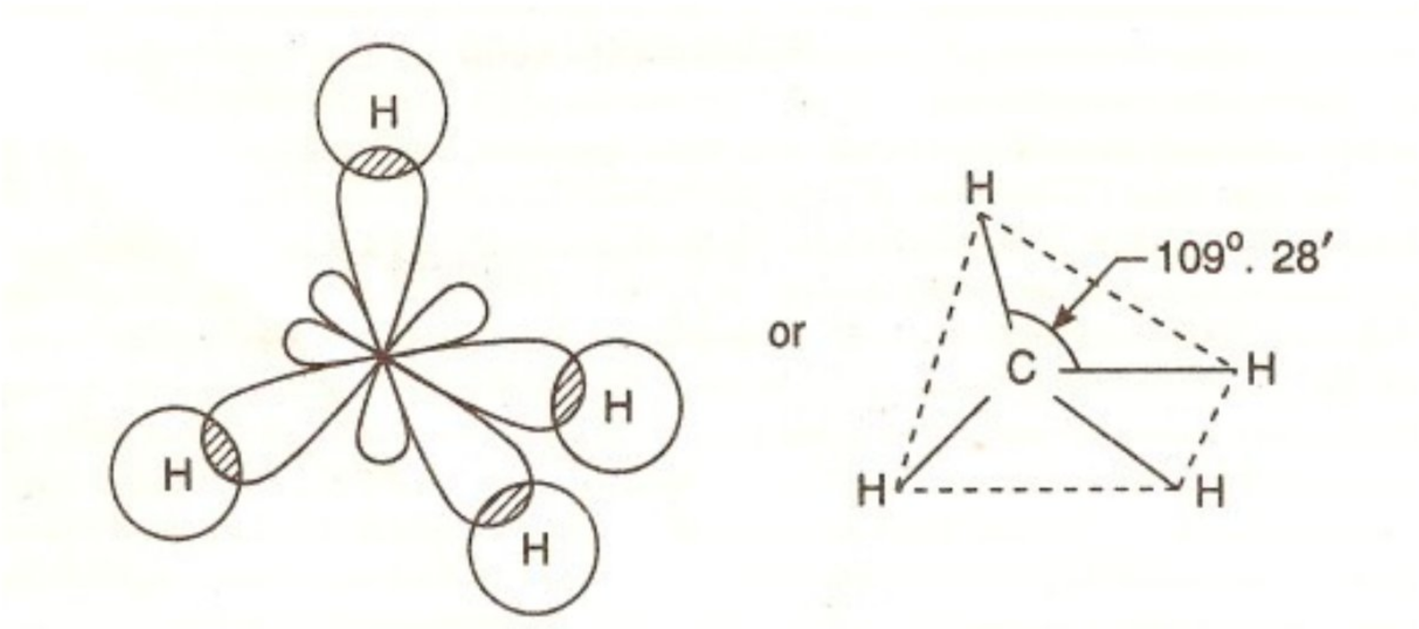
(3) sp3-hybridization: The combination of one s and three p-orbitals to form four hybrid orbitals of equal energy is known as sp3-hybridization.

Example: Methane (CH4) molecule.



HYBRIDIZATION

These sp3-hybridized orbitals are oriented at an angle of 109°28'. When these four sp3 hybrid orbitals overlaps with four 1s orbitals of hydrogen, a symmetrical tetrahedral shaped CH4 molecule form.



VALENCE SHELL ELECTRON PAIR REPULSION (VSEPR) THEORY

Definition: The Valence-Shell-Electrons-Pair-Repulsion Theory (VSEPR), proposes that the stereochemistry of an atom in a molecule is determined primarily by the repulsive interactions among all the electron pairs in its valence shell.

Postulates of VSEPR Theory : The main postulates of this theory are :

(1) The shape of the molecule is determined by repulsion between all of the electron pairs present in the valence shell.

VALENCE SHELL ELECTRON PAIR REPULSION (VSEPR) THEORY

(2) A lone pair of electrons takes up more space round the central atom than a bond pair, since the lone pair is attracted to one nucleus whilst the bond pair is shared by two nuclei. It follows that repulsion between two lone pairs is greater than the repulsion between a lone pair and a bond pair, which in turn is greater than the repulsion between two bond pairs. The repulsive interactions decrease in the order :

LP-LP > LP-BP > BP-BP

(3) The magnitude of repulsion between bonding pairs of electrons. depends on the electronegativity difference between the central atom and the other atoms.

VALENCE SHELL ELECTRON PAIR REPULSION (VSEPR) THEORY

(4) Double bonds cause more repulsion than single bonds, and triple bonds cause more repulsion than a double bond. Repulsive forces decrease sharply with increasing bond angle between the electron pairs.

Example: BF_3 Molecule

In BF_3 the central B atom has the configuration : $1s^2$, $2s^2$, $2p_x^1$. During the bond formation one 2s-electron is promoted to vacant 2py orbital. Thus, excited B (configuration: $1s^2$, $2s^1$, $2P_x^1$, $2p_y^1$, $2p_z^0$) has three unpaired electrons for bond formation with three fluorine atoms. The three bonds between B and F atoms should be slightly different strengths, because in one, 2s; while in other, two 2p-orbital electrons are involved. But in fact, all the three bonds is BF3 are of equal strength with bond angle of 120°.

VSEPR Theory

Used to provide **shape** and **electronic geometry** of covalent compounds.

1. Shape of a molecule depends upon the **number of valence shell electron pairs** around the **central atom**

2. Valence shell is taken as a sphere with the **electron pairs localising** on the **spherical surface**

3. **Electron pairs** in the **valence shell** repel one another since, they are all **negatively charged**

VSEPR Theory

4. Electron pairs occupy positions in space that tend to **minimise repulsion.**

5. **Lone pair** occupies more space on the sphere. So, the order of repulsion is:
lp-lp > lp-bp > bp-bp
(lp: Lone pair,
bp: Bonding pair)

VSEPR Theory

6. A **multiple bond** is treated as a **single bonding pair.**

There is **no effect** of **pi bond** on geometry and shape

4. Formation of NH_3 with sp^3 hybridisation:

N(7) → $1s^2\ \underbrace{2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^1}$ → sp^3 hybridisation

1↓	1	1	1

↳ lone pair

H(1) → $1s^1$

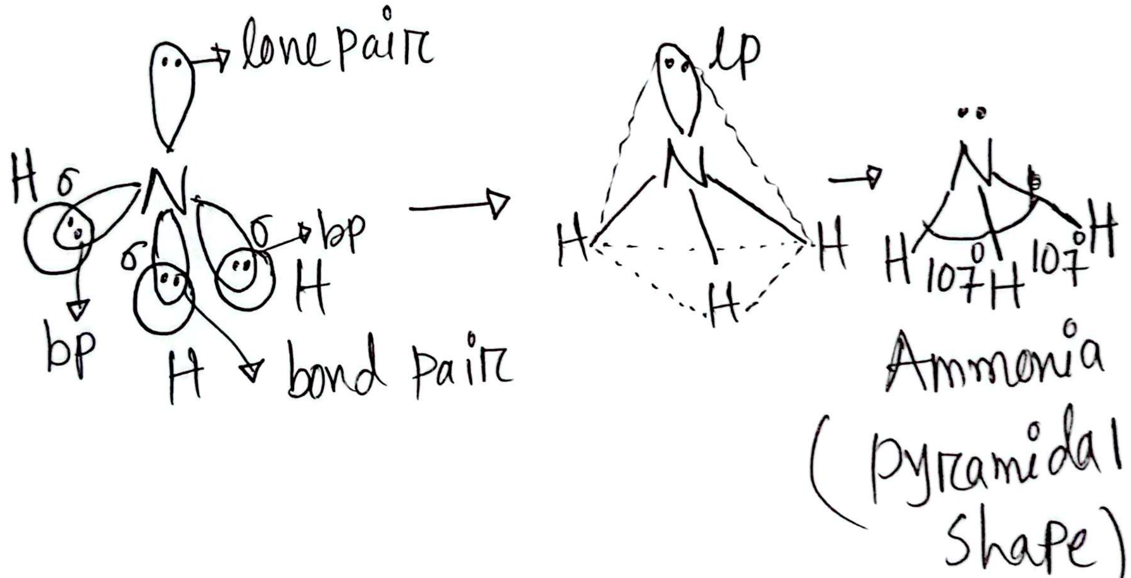


Fig: Orbital Overlaps in the formation of ammonia molecule and formation of tetrahedral geometry of NH_3.

We have seen that the central N-atom has in its valence shell, three bond pairs (bP) and one lone pair (lP) of electrons.

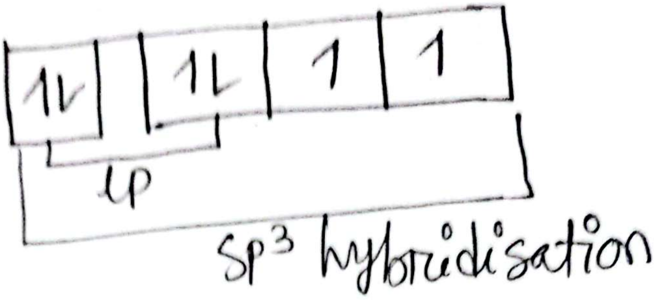
The three bond pairs and one lone pair may get arranged tetrahedrally with the central atom. The central atom exercise different pulls on them.

According to the VSEPR theory lp-bp show greater repulsion than bp-bp repulsion. Here lp-bp have greater repulsive force than bp-bp repulsive force.

As a result HNH bond angles is 107° rather than the regular tetrahedral angle of 109.5° for sp^3 hybridisation. And shape of NH_3 molecule form Pyramidal structure rather than tetrahedral structure of sp^3 hybridisation.

⇒ 5. Formation of H_2O with sp^3 hybridisation:

$O(8) \rightarrow 1s^2\ 2s^2\ 2p_x^2\ 2p_y^1\ 2p_z^1$



Hybrid state:

$O(8) \rightarrow 1s^2\ 2\psi_1^2\ 2\psi_2^2\ 2\psi_3^1\ 2\psi_4^1$

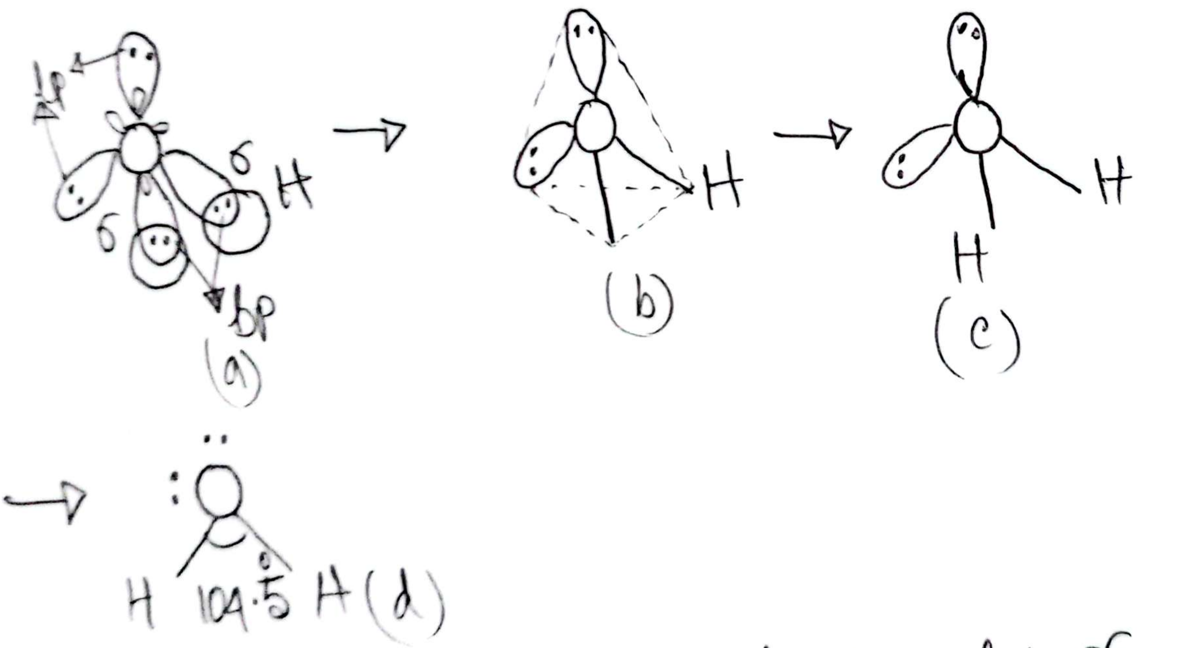


Fig: Formation of tetrahedral model of water molecule showing two lone pairs at O-atom (in Fig a & b) Showing lp-lp, bp-bp and lp-bp repulsions (in Fig c & d)

We have seen that the central O- atom has in its valence shell, two bond pairs (bp) and two lone pairs.

Here, lp-lp repulsion > lp- bp repulsion > bp-bp repulsion is happen according to the VSEPR theory. lp-lp electrons show greater repulsion than lp- bp and bp- bp repulsion.

As a result, HOH bond angles is 104.5° rather than the regular tetrahedral angle of 109.5° for sp^3 hybridisation.

And shape of H_2O molecule form V-shape structure rather than the tetrahedral structure of sp^3 hybridisation.

MOLECULAR ORBITAL THEORY

Introduction: The molecular orbital theory given by Hund and Mullikens considers that the valency electrons are associated with all the concerned nuclii, *i.e.,* with the molecule as a whole. Thus, in other words, the atomic orbitals combine together to form molecular orbitals in which the identity of both the atomic orbitals is lost.

Postulates of Molecular Orbital Theory :

(a) During the formation of bond, the two half-filled atomic·orbitals with opposite spins combine to form two molecular orbitals (M.O.). One of which is bonding M.O. and other antibonding M.O. The energy of combining atomic orbitals, however, should be of similar magnitude.

MOLECULAR ORBITAL THEORY

(b) A bonding M.O. has lower energy than the atomic orbitals from which it is formed; while antibonding M.O. has higher energy than the atomic orbitals from which it is formed.

(c) Two *s* or p-orbitals can overlap end-to-end to form one sigma bonding molecular orbital (σ) and one sigma antibonding molecular orbital (σ *).

(d) Two p-orbitals can overlap sideways to form one pi (π)-bonding molecular orbital and pi (π *) anti bonding orbital.

(e) The energies of σ -bonding orbitals are lower than those of π -bonding orbitals.

(f) The orbitals of lower energy are filled first. Each orbital may hold up two electrons, provided that they have opposite spins.

MOLECULAR ORBITAL THEORY

(g) When several orbitals have the same energy, *(i.e.,* they are degenerate), electrons will be arranged so as to give the maximum number of unpaired spins (Hund's rule).

MOLECULAR ORBITAL THEORY

(g) Molecules or ions having one or more unpaired electrons in the M.O. are paramagnitic in nature; while those with paired electron in the M.O., are diamagnitic in nature.

Bond order : Bond order of a molecule is :

$$= \frac{\begin{bmatrix} \text{No. of electrons in the} \\ \text{bonding molecular orbitals} \end{bmatrix} - \begin{bmatrix} \text{No. of electrons in anti-} \\ \text{bonding molecular orbitals} \end{bmatrix}}{2} = \frac{(Nb - Na)}{2}$$

Combination of Atomic Orbitals

(i) **Combinaltion of s-orbitals:** The combination of two similar 1*s* atomic orbitals gives rise to two molecular orbitals, σ1s and σ *1s. σ1s is the bonding molecular orbital, whereas σ *1s is the antibonding molecular orbital. Similarly, the combination of two 2s atomic orbitals will give rise to σ2s and σ*2s molecular orbitals. These will be of higher energy than σ1s and σ*1s molecular orbitals as they are formed from atomic orbitals of higher energy.

(ii) **Combination of p-orbitals :** The p-orbitals can combine either along the axis to give σ molecular orbitals or perpendicular to the axis to give π-molecular orbitals.

Combination of Atomic Orbitals

(a) Combination of p_x orbitals: The overlapping of two P_x orbitals along the axis results in the formation of a bonding σ_{px} and antibonding σ^*_{px} molecular orbitals. The orbitals with same sign produce bonding molecular orbitals, whereas the orbitals with unlike signs produce anti bonding molecular orbitals .

(b) Combination of p_y or p_z orbitals: The overlapping of two p_y or two p_z atomic orbitals takes place perpendicular to the molecular axis and, thus, results in the formation of bonding and antibonding π- molecu!ar orbitals.